ECE250: Signals & Systems
End-Semester Exam

Date: 09th December, 2025 **Total Marks:** 38 **Duration:** 2.5 hour

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Note:

1. Please provide proper mathematical justifications with your answers. No marks will be awarded without a valid justification.
2. Institute plagiarism policy are strictly applicable.

[Q1] [CO1, CO2]: Determine and sketch the convolution of the following two signals: **[3 Marks]**

$$x(t) = \begin{cases} t+1, & 0 \le t \le 1 \\ 2-t, & 1 < t \le 2 \\ 0, & \text{elsewhere} \end{cases}$$

$$h(t) = \delta(t+2) + 2\delta(t+1).$$

[Q2] [CO2]: Let $x(t)$ be a periodic signal with fundamental period $T = 2$ and Fourier coefficients a_k.

$$x(t) = \begin{cases} t, & 0 \le t \le 1 \\ 2-t, & 1 \le t \le 2 \end{cases}$$

(a) Determine the value of a_0. **[1 Mark]**

(b) Determine the Fourier series representation of $dx(t)/dt$. **[2 Marks]**

(c) Use the result of part (b) and the differentiation property of the continuous-time Fourier series to determine the Fourier series coefficients of $x(t)$. **[2 Marks]**

[Q3] [CO2, CO4]: Consider a causal LTI system with frequency response

$$H(j\omega) = \frac{1}{j\omega + 3}.$$

For a particular input $x(t)$, this system is observed to produce the output

$$y(t) = e^{-3t}u(t) - e^{-4t}u(t).$$

Determine $x(t)$. **[6 Marks]**

[Q4] [CO1, CO3, CO5]: Let $x[n]$ be a discrete-time signal with Fourier transform $X(e^{j\omega})$, which is illustrated in Figure 1.

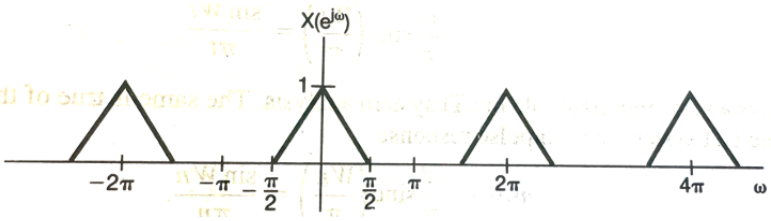


Figure 1

Sketch the Fourier transform of $v[n] = x[n]p[n]$ for each of the following signals $p[n]$:

(i) $p[n] = \cos \pi n$ **[3 Marks]**

(ii) $p[n] = \sum_{k=-\infty}^{\infty} \delta[n - 2k]$ **[3 Marks]**

[Q5] [CO4]: Suppose the following facts are given about the signal $x(t)$ with Laplace transform $X(s)$: **[6 Marks]**

1. $x(t)$ is real and even.
2. $X(s)$ has four poles and no zeros in the finite s-plane.
3. $X(s)$ has a pole at $s = (1/2)e^{j\pi/4}$.
4. $\int_{-\infty}^{\infty} x(t)dt = 4$.

Determine $X(s)$ and its ROC.

[Q6] [CO1, CO2, CO5]: The input $x[n]$ and output $y[n]$ of a LTI system are related to the block diagram representation shown below in Figure-2.

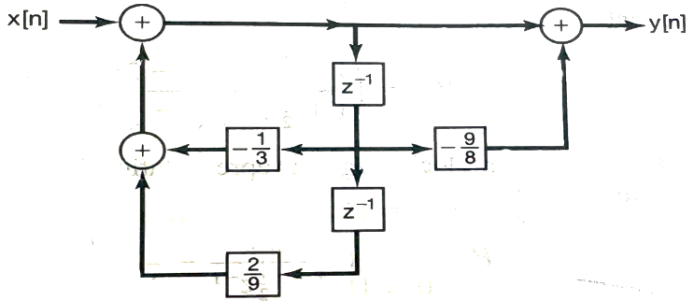


Figure 2

(a) Determine the difference equation relating $y[n]$ and $x[n]$. **[2 Marks]**

(b) Find and plot poles and zeros in the z-plane. **[1 Mark]**

(c) Plot all possible ROCs. **[1 Mark]**

(d) Identify the ROC that can lead to a causal and stable system. Correspondingly, find $h[n]$. **[2 Marks]**

[Q7] [CO1, CO2, CO3]:

(a) Write the Nyquist sampling theorem. **[2 Marks]**

(b) In the system shown in Figure-3, two functions of time, $x_1(t)$ and $x_2(t)$, are multiplied together, and the product $v(t)$ is sampled by a periodic impulse train. $x_1(t)$ is band limited to ω_1, and $x_2(t)$ is band limited to ω_2, that is, **[4 Marks]**

$$X_1(j\omega) = 0, |\omega| \geq \omega_1,$$

$$X_2(j\omega) = 0, |\omega| \geq \omega_2.$$

Determine the maximum sampling interval T such that $v(t)$ is recoverable from $v_p(t)$ through the use of an ideal lowpass filter.

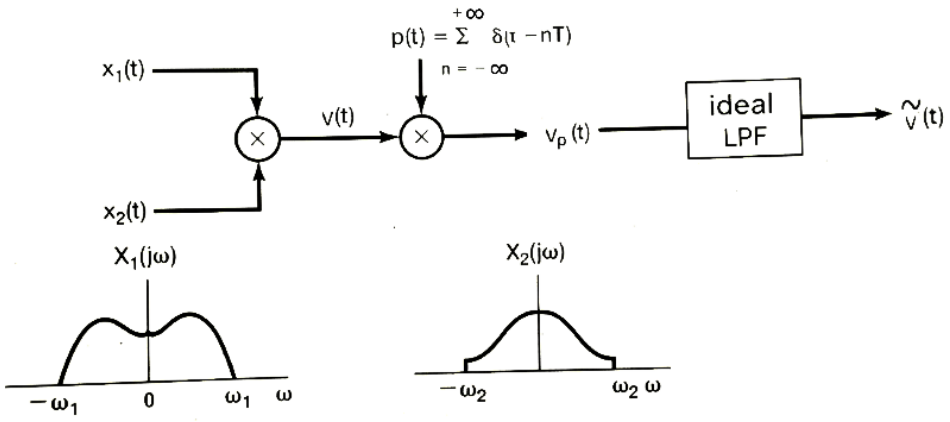


Figure 3

(c) **[BONUS]** Let

$$x_1(t) = \cos(\omega_1 t),$$
$$x_2(t) = \cos(\omega_2 t).$$

with $\omega_1 = 2\pi f_1$ where $f_1 = 2$kHz, and $\omega_2 = 2\pi f_2$ where $f_2 = 5$kHz

Show the frequency domain spectrum of $X_1(j\omega)$, $X_2(j\omega)$, $V_p(j\omega)$,$\tilde{V}(j\omega)$, and the bandwidth of ideal lowpass filter corresponding to max sampling interval T such that $v(t)$ is recoverable from $v_p(t)$ through the use of an ideal lowpass filter. **[0.5x5=2.5 Marks]**

(d) **[BONUS]** If T = 10^{-4}, starting from the spectrum of $V_p(j\omega)$, show the bandwidth/spectrum of ideal lowpass filter, draw the frequency domain spectrum of $\tilde{V}(j\omega)$, and write the time domain expression of $\tilde{v}(t)$. **[0.5+1.5+1.5=3.5 Marks]**